



Yakeen Plus (2025)

SHORT PRACTICE TEST - 02

DURATION : 60 Minutes

DATE : 18/09/2024

M. MARKS : 192

ANSWER KEY

PHYSICS

1. (1)
2. (4)
3. (4)
4. (2)
5. (1)
6. (3)
7. (3)
8. (2)
9. (1)
10. (2)
11. (4)
12. (2)

CHEMISTRY

13. (3)
14. (3)
15. (3)
16. (1)
17. (3)
18. (2)
19. (4)
20. (4)
21. (1)
22. (1)
23. (3)
24. (3)

BOTANY

25. (1)
26. (2)
27. (4)
28. (4)
29. (3)
30. (1)
31. (2)
32. (2)
33. (4)
34. (2)
35. (2)
36. (3)

ZOOLOGY

37. (3)
38. (1)
39. (3)
40. (1)
41. (3)
42. (4)
43. (1)
44. (2)
45. (3)
46. (3)
47. (2)
48. (3)

SECTION-(I) PHYSICS

1. (1)

$$\text{Stopping distance} = \frac{u^2}{2a}$$

[New NCERT Class 11th Page No. 21]

2. (4)

$$x = ae^{-\alpha t} + be^{\beta t}$$

$$\frac{dx}{dt} = a(-\alpha)e^{-\alpha t} + b(\beta)e^{\beta t}$$

$$v = b\beta e^{\beta t} - a\alpha e^{-\alpha t}$$

As we increase time $e^{\beta t}$ increases and $e^{-\alpha t}$ decreases. So, v keeps on increasing with time.

[New NCERT Class 11th Page No. 11]

3. (4)

$$t_1 = \sqrt{\frac{2\left(\frac{h}{4}\right)}{g}}$$

$$t_2 = \sqrt{\frac{2\left(4 \cdot \frac{h}{4}\right)}{g}} - \sqrt{\frac{2\left(3 \cdot \frac{h}{4}\right)}{g}}$$

$$\text{Now, } \frac{t_2}{t_1} = \frac{(2-\sqrt{3})\sqrt{\frac{2h}{g}}}{\sqrt{\frac{2h}{g}}} \Rightarrow \frac{t_2}{t_1} = \frac{2-\sqrt{3}}{1}$$

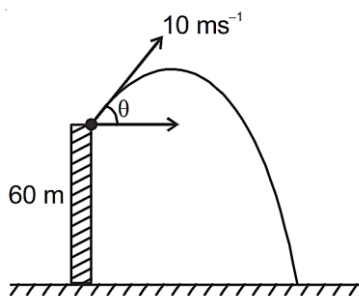
[New NCERT Class 11th Page No. 20]

4. (2)

$$T = \frac{2m_1m_2g}{m_1+m_2} = \frac{2 \times 3 \times 6 \times 10}{(3+6)} = 40 \text{ N}$$

[New NCERT Class 11th Page No. 59]

5. (1)



$$s_y = u_y T + \frac{1}{2} a_y T^2$$

$$-60 = 10 \sin 30^\circ \cdot T - \frac{1}{2} g T^2$$

$$-60 = 5T - 5T^2$$

$$T^2 - T - 12 = 0$$

$$T = 4 \text{ s}$$

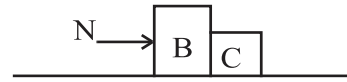
[New NCERT Class 11th Page No. 40]

6. (3)

$$F = ma$$

$$14 = (4 + 2 + 1)a$$

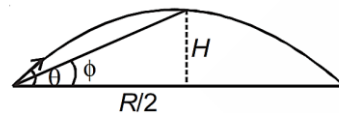
$$a = 2 \text{ m/s}^2$$



$$N = (2 + 1) \times 2 = 6$$

[New NCERT Class 11th Page No. 58]

7. (3)



$$\tan \phi = \frac{H}{R/2}$$

$$R = 4H \cot \theta$$

$$\tan \phi = \frac{2H}{4H \cot \theta}$$

$$\tan \phi = \frac{1}{2} \tan \theta$$

[New NCERT Class 11th Page No. 40]

8. (2)

$$\text{Also, } t_1 + t_2 = \frac{2u \sin \theta}{g}$$

$$4 = \frac{2 \times 40 \times \sin \theta}{10}$$

$$\sin \theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

$$\text{So, } \tan \theta = \tan 30^\circ \Rightarrow \frac{1}{\sqrt{3}}$$

$$\theta = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

[New NCERT Class 11th Page No. 40]

9. (1)

$$v^2 = u^2 + 2as$$

$$\Rightarrow (-4u)^2 = u^2 + 2(-g)(-h)$$

$$\Rightarrow 16u^2 = u^2 + 2gh$$

$$\Rightarrow 15u^2 = 2gh$$

$$\Rightarrow h = \frac{15u^2}{2g}$$

[New NCERT Class 11th Page No. 18, 19]

10. (2)

The object starts from the rest

$$\text{Distance travelled in last second} = \frac{9h}{25}$$

Let n is the total time

$$h = 0 + \frac{1}{2}gn^2 \quad \dots(i)$$

Distance travelled in $(n-1)$ second

$$= h - \frac{9h}{25} = \frac{16h}{25}$$

$$\frac{16h}{25} = \frac{1}{2}g(n-1)^2$$

By using value of h from eqn. (1) in eqn. (2)

$$\frac{16}{25} \frac{gn^2}{2} = \frac{1}{2}g(n-1)^2$$

Taking under root both side

$$\frac{4}{5}n = n-1$$

$$\frac{n}{5} = 1$$

$$n = 5$$

$$\text{Now, } h = \frac{1}{2} \times 9.8 \times 5^2 = 122.5 \text{ m}$$

[New NCERT Class 11th Page No. 17, 18]

11. (4)

$$\frac{\text{Total distance}}{\text{Total time}} = \frac{70}{5} = 14 \text{ m/s}$$

[New NCERT Class 11th Page No. 14, 15]

12. (2)

$$(A) \vec{C} - \vec{A} - \vec{B} = 0$$

$$\vec{A} + \vec{B} = \vec{C}$$

$$(B) \vec{A} - \vec{C} - \vec{B} = 0$$

$$\vec{C} + \vec{B} = \vec{A}$$

$$(C) \vec{B} - \vec{A} - \vec{C} = 0$$

$$\vec{A} + \vec{C} = \vec{B}$$

$$(D) \vec{A} + \vec{B} = -\vec{C}$$

$$\vec{A} + \vec{B} + \vec{C} = 0$$

A-IV, B-III, C-I, D-II

[NCERT Class 11th, Page No.68, 69]

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SECTION-(II) CHEMISTRY

13. (3)

$$n_B = \frac{3}{60} = 0.05; n_A = \frac{45}{18} = 2.5$$

$$\frac{\Delta p}{p^0} = \chi_B = \frac{0.05}{2.5 + 0.05} = 0.0196 = 0.02$$

[New NCERT Class 12th Page No. 4]

14. (3)

$$\Delta T = K_f \times \frac{w_B \times 1000}{M_B \times w_A}$$

$$0.186 = 1.86 \times \frac{w_B \times 1000}{60 \times 500}$$

$$w_B = 3g$$

[New NCERT Class 12th Page No. 19]

15. (3)

$$\pi = iCRT$$

Greater the no. of particles, greater the osmotic pressure.

[New NCERT Class 12th Page No. 21]

16. (1)

For isotonic solutions:

$$\pi_1 = \pi_2 \quad \therefore \pi = CRT$$

$$C_1 = C_2$$

$$\frac{1.5/60}{V} = \frac{5.25/M}{V}$$

$$\therefore M = 210 \text{ g/mol}$$

[New NCERT Class 12th Page No. 21]

17. (3)

Volume of solution

$$= \frac{\text{Mass}}{\text{Density}} = \frac{1120}{1.15} = 973.9 \text{ mL}$$

$$\text{Molarity, } M = \frac{w_B \times 1000}{M_B \times V}$$

$$= \frac{120 \times 1000}{60 \times 973.9} = 2.05M$$

[New NCERT Class 12th Page No. 20]

18. (2)

Ethanol + H₂O → Azeotropic mixture, Non-ideal solution with positive deviation.

Ebullioscopic constant (K_b) ⇒ Molal elevation constant.

Cryoscopic constant (K_f) ⇒ Molal depression constant

H₂O + H₂SO₄ → Azeotropic mixture, Non-ideal solution with negative deviation.

[New NCERT Class 12th Page No. 18]

19. (4)
For a solution exhibiting positive deviation from Raoult's law, A — B attraction forces < A — A and B — B attraction forces.
[New NCERT Class 12th Page No. 16]

20. (4)
For NH_4Cl , $i = 2$ which is lowest among the given solutes. Hence depression in freezing point will be least for it.
 $\therefore$ Highest freezing point of solution.
[New NCERT Class 12th Page No. 14]

21. (1)
$$P_T = P_{\text{benzene}}^\circ \chi_{\text{benzene}} + P_{\text{toluene}}^\circ \chi_{\text{toluene}}$$
$$760 = 957 \cdot \chi_{\text{benzene}} + 379.5(1 - \chi_{\text{benzene}})$$
$$\chi_{\text{benzene}} = 0.66 \text{ and } \chi_{\text{toluene}} = 1 - 0.66 = 0.34$$

[New NCERT Class 12th Page No. 12]

22. (1)
At constant T,
 $P_1 V_1 = P_2 V_2$
 $1 \times 20 = P_2 \times 50$; $P_2 = \frac{20}{50} \times 1$
[New NCERT Class 11th Page No. 143]

23. (3)
The value of universal gas constant can be expressed in different units and its value would depend only on the units of the measurement.
[New NCERT Class 11th Page No. 144]

24. (3)
 $V_1 = 100 \text{ mL}$ $V_2 = 80 \text{ mL}$
 $T_1 = 300 \text{ K}$ $T_2 = ?$
 $P_1 = 740 \text{ mm of Hg}$ $P_2 = 740 \text{ mm of Hg}$
Applying Charles law $V \propto T$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{100}{300} = \frac{80}{T_2}$$

$$T_2 = \frac{300 \times 80}{100} = 240 \text{ K}$$

$$T_2 = 240 - 273 = -33^\circ\text{C}.$$

[New NCERT Class 11th Page No. 144]

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SECTION-(III) BOTANY

25. (1)
In an animal cell, cytokinesis is achieved by the appearance of a furrow in the plasma membrane. The furrow gradually deepens and ultimately joins in the centre dividing the cell cytoplasm into two.
[New NCERT Class 11th Page No. 124]

26. (2)
- | | |
|--|------------|
| Chromosomes start pairing | Zygotene |
| Four chromatids of each bivalent appear as tetrads | Pachytene |
| Formation of X-shaped structure chiasmata | Diplotene |
| Transition to metaphase | Diakinesis |
- [New NCERT Class 11th Page No. 126]

27. (4)
Gametes are formed from specialised diploid cells. Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organisms. During meiosis, chromosome number reduces by half and results in the production of haploid daughter cells.
[New NCERT Class 11th Page No. 125]

28. (4)
The growth of multicellular organisms is due to mitosis. Cell growth results in disturbing the ratio between the nucleus and the cytoplasm. It therefore becomes essential for the cell to divide to restore the nucleo-cytoplasmic ratio. A very significant contribution of mitosis is cell repair.
[New NCERT Class 11th Page No. 125]

29. (3)
Some cells in the adult animals do not appear to exhibit division (e.g., heart cells) and many other cells divide only occasionally, as needed to replace cells that have been lost because of injury or cell death. These cells that do not divide further exit G_1 phase to enter an inactive stage called quiescent stage (G_0) of the cell cycle.
[New NCERT Class 11th Page No. 122]

30. (1)
Four haploid cells are formed at the end of meiosis II.
[New NCERT Class 11th Page No. 124, 125]

31. (2)
The complete disintegration of the nuclear envelope marks the start of the second phase of mitosis, hence the chromosomes are spread through the cytoplasm of the cell.

[New NCERT Class 11th Page No. 123]

32. (2)
Anaphase II: It begins with the simultaneous splitting of the centromere of each chromosome (which was holding the sister chromatids together), allowing them to move toward opposite poles of the cell by shortening of microtubules attached to kinetochores.

[New NCERT Class 11th Page No. 127]

33. (4)
(a) – Pre-mitotic phase, (b) – M phase, (c) – post mitotic phase, (d) – G₁ Stage, (e) – S phase.

[New NCERT Class 11th Page No. 121]

34. (2)
The number of chromosome and amount of DNA in maize root tip cell in G₁ phase of cell cycle is 20 and 20 Pg respectively.

After S phase, the number of chromosome remain same = 20 and the amount of DNA doubles = 40pg.

[New NCERT Class 11th Page No. 121]

35. (2)
Nucleolus, golgi complex and ER reforms in telophase.
Cells are metabolically active but no longer proliferate in G₀ stage.
Centromere split and chromatids separate during anaphase.

[New NCERT Class 11th Page No. 122, 123, 124]

36. (3)
S or synthesis phase marks the period during which DNA synthesis or replication takes place. During this time the amount of DNA per cell doubles. If the initial amount of DNA is denoted as 2C then it increases to 4C. However, there is no increase in the chromosome number; if the cell had diploid or 2n number of chromosomes at G₁, even after S phase the number of chromosomes remains the same, i.e., 2n.

[New NCERT Class 11th Page No. 121]

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SECTION-(IV) ZOOLOGY

37. (3)
Every 100 ml of deoxygenated blood delivers approximately 4 ml of CO₂ to alveoli.

[New NCERT Class 11th Page No. 190]

38. (1)
Nearly 20-25 per cent of CO₂ is transported by RBCs whereas 70 per cent of it is carried as bicarbonate. About 7 per cent of CO₂ is carried in a dissolved state through plasma.

[New NCERT Class 11th Page No. 189]

39. (3)
Trachea divides at the level of 5th thoracic vertebrae.

[New NCERT Class 11th Page No. 184]

40. (1)

Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated)	Tissues
O ₂	159	104	40	95	40
CO ₂	0.3	40	45	40	45

[New NCERT Class 11th Page No. 187]

41. (3)
• Volumes of air involved in breathing movement can be estimated with the help of spirometer.
• Each haemoglobin molecule can carry a maximum of four molecules of O₂.

[New NCERT Class 11th Page No. 191]

42. (4)
Birds respire through lungs.

[New NCERT Class 11th Page No. 183]

43. (1)
• Expiration is a process from which the alveolar air is released out.
• It is due to the high intra pulmonary pressure present in the lungs.

[New NCERT Class 11th Page No. 185]

44. (2)
• Long exposure can give rise to inflammation leading to fibrosis proliferation of fibrous tissues and thus causing serious lung damage.
• Emphysema is a chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased.

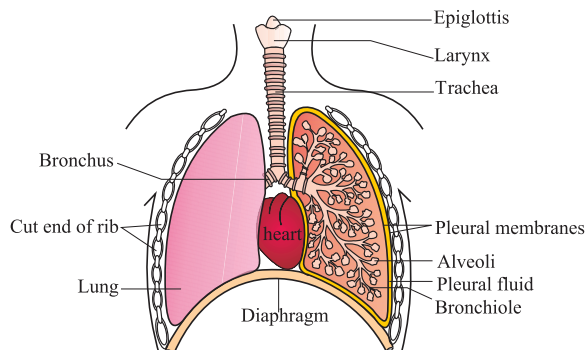
[New NCERT Class 11th Page No. 190, 191]

45. (3)

Inspiratory Capacity	3000-3500 ml
Expiratory Capacity	1500-1600 ml
Functional Residual Capacity	2100-2200 ml
Vital Capacity	4000-4600 ml

[New NCERT Class 11th Page No. 187]

46. (3)



Alveoli is a thin walled vascular bag like structure and it is the site for the exchange of gases.

[New NCERT Class 11th Page No. 184]

47. (2)

- Pneumotaxic centre is present in the pons region of brain.
- About 20-25 per cent CO₂ is carried by haemoglobin as carbamino-haemoglobin

[New NCERT Class 11th Page No. 190]

48. (3)

In the alveoli, where there is high pO₂, low CO₂, lesser H⁺ concentration and lower temperature, the factors are all favourable for the formation of oxyhaemoglobin.

[New NCERT Class 11th Page No. 189]

